

1. MACHINE IDENTIFICATION

The Press-2001 is a 1500-ton mechanical (flywheel) press. It has NO hydraulic system ? clamping force is generated by a motor-driven flywheel and eccentric gear. Serial prefix MP- identifies the mechanical press family. Do not apply Press-2000 hydraulic error codes to this machine.

1.1 LUBRICATION SYSTEM

The gearbox and clutch are splash-lubricated with ISO-VG220 gear oil. The reservoir holds 6 liters. Change oil every 3000 operating hours.

The slide guides are grease-lubricated. Grease fittings are located on the front and rear of each guide rail. Apply lithium-based grease (NLGI 2) weekly.

1.2 CLUTCH/BRAKE UNIT

The press uses a pneumatically actuated clutch and brake unit. Air supply must be at 6-8 bar, dry and filtered. The clutch engages when the solenoid valve opens, releasing the spring-applied brake. Brake wear is indicated by increased stopping time. Measure stopping angle annually ? should not exceed 30°.

7. TROUBLESHOOTING

7.1 E101 ? Drive Chain Binding

Meaning: The main drive chain tension is excessive or the chain is catching on the guard. The controller detects increased torque on the flywheel motor.

NOTE: E101 on the Press-2001 means DRIVE CHAIN BINDING, not winding temperature (RoboInject-300) and not low hydraulic pressure (Press-2000). Always verify the machine model before diagnosing.

Probable causes:

- Drive chain tension too high (overtightened)
- Chain sprocket misalignment between motor and flywheel
- Worn chain links causing the chain to 'skip' under load
- Foreign object caught in the chain guard
- Flywheel bearing seizure

Corrective action:

Step 1. Remove the chain guard and inspect the chain condition visually.

Step 2. Measure chain tension at the center of the longest span. Deflection should be 15-20mm when pressed with 5 kg force.

Step 3. Check sprocket alignment: use a straightedge across both sprockets. Misalignment >2mm causes accelerated wear.

Step 4. Lubricate the chain with ISO-VG320 oil. Run the motor without load for 5 minutes to distribute oil.

Step 5. If binding persists, replace the chain set (P/N CH-2001-KIT).

Source: Section 7, page 311.

7.2 E204 ? Flywheel Overspeed

Meaning: The flywheel speed has exceeded 110% of the rated max. The VFD has applied the dynamic brake.

Probable cause: VFD parameters misconfigured or feedback encoder fault.

7.3 E312 ? Clutch Air Pressure Low

Meaning: The air supply to the clutch solenoid has dropped below 5 bar. The press will not cycle.

Corrective action:

Step 1. Check the main air supply pressure at the regulator.

Step 2. Inspect the air line filter ? drain if water is present.

Step 3. Test the solenoid valve by applying 24V DC directly. If the valve clicks but no air flows, replace the valve coil (P/N V-SOL-2001).

8. PERIODIC MAINTENANCE

Daily: visual check of chain tension, listen for abnormal gear noise, check air pressure on regulator gauge.

Weekly: grease guide rails, drain air line filter, inspect clutch brake pads.

Monthly: check gear oil level, inspect flywheel bearing temperature (touch test).

Annually: change gearbox oil, replace clutch brake pads, full electrical inspection.